



Hands-On Lab Exercise 12: DevOps Automation Using Shell Script

Topic: Real-World DevOps Automation Scenarios

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DevOps automation means using scripts and tools to remove manual work from system operations. In real IT environments, tasks such as system setup, package installation, user creation, service management, log handling, backup creation, monitoring, and reporting are done every day. If these tasks are done manually, they consume time, introduce human errors, and create inconsistency.

Shell scripting plays a very important role in DevOps because it allows automation at the operating system level. It connects infrastructure, applications, services, and users. A shell script can configure systems, deploy applications, manage services, monitor resources, and maintain infrastructure automatically.

DevOps automation is not about one script — it is about building systems of scripts that work together. These scripts create repeatable processes, stable systems, and reliable operations. This is how modern infrastructure and cloud platforms are managed.

In simple words, DevOps automation means letting systems manage themselves using scripts.



Learning Objective

- Understand DevOps automation
 - Learn how shell scripts support DevOps
 - Learn real automation use cases
 - Build integrated automation scripts
 - Understand system-level automation
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Learning Outcome

After this lab, participants will: - Build DevOps automation scripts - Automate infrastructure tasks - Automate service management - Automate deployment tasks - Understand real-world DevOps scripting



Example 1: System Setup Automation Script



Purpose:

Automatically prepare a new server

```
#!/bin/bash

echo "Starting system setup..."

echo "Updating system..."
apt update -y || yum update -y

echo "Installing basic packages..."
apt install -y git curl wget || yum install -y git curl wget

echo "Creating directories..."
mkdir -p /opt/devops/apps
mkdir -p /opt/devops/logs

echo "Setup completed"
```



Example 2: User and Permission Automation

```
#!/bin/bash

echo "Creating DevOps user..."
useradd devops

echo "Setting password..."
echo "devops:Devops@123" | chpasswd

echo "Giving sudo access..."
usermod -aG sudo devops || usermod -aG wheel devops

echo "User setup completed"
```



Example 3: Application Deployment Automation

```
#!/bin/bash

app_dir=/opt/devops/apps/myapp

mkdir -p $app_dir

cd $app_dir

echo "Deploying application..."
touch app.sh

echo "#!/bin/bash" > app.sh
echo "echo Application Running" >> app.sh

chmod +x app.sh

./app.sh
```



Example 4: Service Management Automation

```
#!/bin/bash

service_name=ssh

echo "Checking service: $service_name"

systemctl start $service_name
systemctl enable $service_name

systemctl status $service_name
```



Example 5: Integrated DevOps Automation Script

```
#!/bin/bash

# Directory setup
mkdir -p /opt/devops/project

# User setup
useradd devopsuser 2>/dev/null

# Package installation
apt install -y git || yum install -y git

# Service setup
systemctl start ssh
systemctl enable ssh

# Application folder
mkdir -p /opt/devops/project/app

# Logging
log_file=/opt/devops/project/setup.log

echo "DevOps automation started" > $log_file
date >> $log_file

echo "System prepared successfully" >> $log_file

cat $log_file
```

Understanding the Flow

Multiple automation steps are combined into one script. This is how real DevOps automation systems are built. Each script becomes a module in a bigger automation architecture.

Key Understanding

DevOps automation is about reliability, repeatability, and scalability. Scripts replace manual operations.

Summary

- Automation removes manual work
 - Scripts create stable systems
 - DevOps needs automation
 - Shell scripting is foundation
 - Real infrastructure uses scripts
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End of Lab

This lab represents **real DevOps automation foundations**.

Now participants understand: - Infrastructure automation - User automation - Deployment automation - Service automation - System integration

This is the bridge between **learning shell scripting** and **real DevOps engineering**.

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